Lab 8 - Edge Detection

In [1]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt

imgFile = 'kid1.jpg'
```

Sobel Edge Detector

In [2]:
```
# 1. Load the real image

img = cv2.imread(imgFile)

# 2. Convert to Grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# 3. Apply Gaussian Blur to reduce noise
# (5, 5) is the kernel size; 0 means standard deviation is calculated from kernel s

# ---------| You can try Sobel with Original Image, Gray scale and Blur
# ---------| Try these
# ---(1)---|  blurred = img
# ---(2)---|  blurred = gray

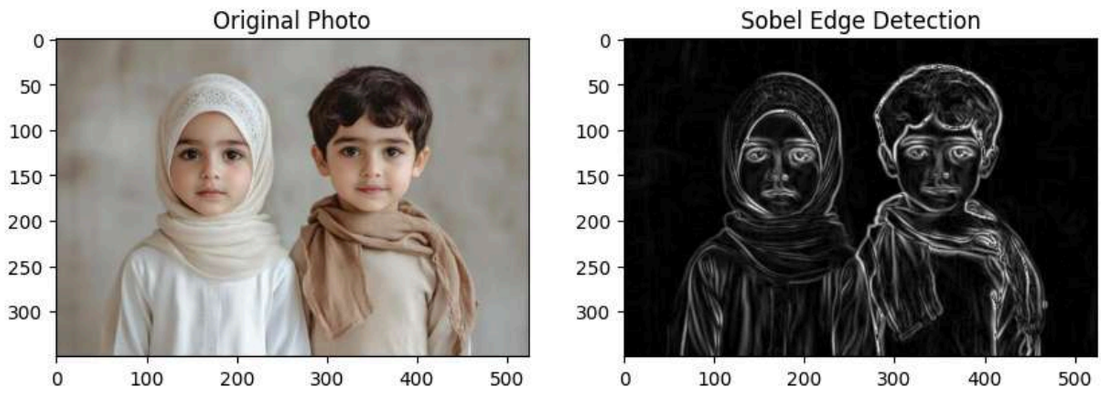
blurred = cv2.GaussianBlur(gray, (5, 5), 0) # blurred = img or gray   ## Check the

# 4. Apply Sobel filter
# cv2.CV_64F handles the high precision needed for derivatives
sobelx = cv2.Sobel(blurred, cv2.CV_64F, 1, 0, ksize=3)
sobely = cv2.Sobel(blurred, cv2.CV_64F, 0, 1, ksize=3)

# 5. Calculate Magnitude (Final Edge Map)
sobel_combined = cv2.magnitude(sobelx, sobely)

# 6. Normalize to 0-255 for display
sobel_final = np.uint8(np.absolute(sobel_combined))

# Visualization
plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1), plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB)), plt.title('
plt.subplot(1, 2, 2), plt.imshow(sobel_final, cmap='gray'), plt.title('Sobel Edge D
plt.show()
```



Robert Edge Detector

Comparing to Sobel

Roberts: Kernel Size --> 2x2 Sobel: Kernel Size --> 3x3

Roberts: Speed --> Faster Sobel: Speed --> Slower

Roberts: Noise Resistance --> Poor (High Senstivity) Sobel: Noise Resistnace --> Good (Built-in Smoothing)

Roberts: Edge Width --> Very thin/sharp Sobel: Edge Width --> Thicker

In [3]:
```
# 1. Load a real image in grayscale
image = cv2.imread(imgFile, 0) #cv2.IMREAD_GRAYSCALE)

# 2. Pre-process: Always blur for Roberts!
# Its 2x2 kernel is very sensitive to noise.
blurred = cv2.GaussianBlur(image, (5, 5), 0)

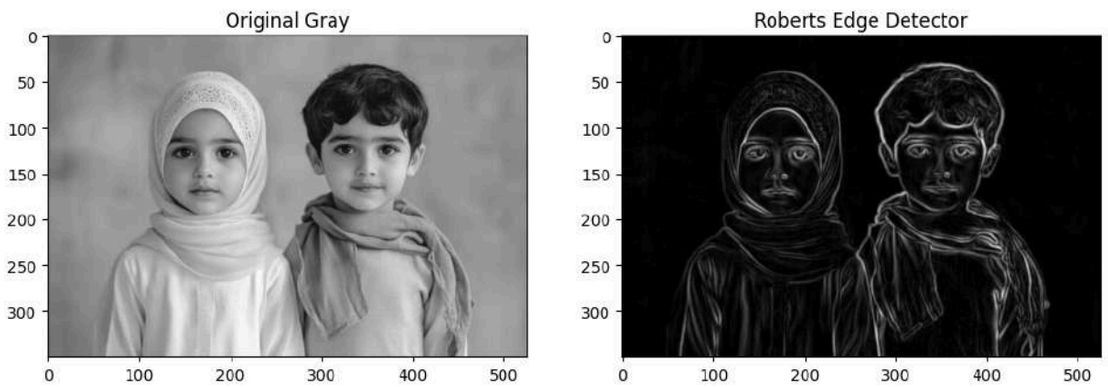
# 3. Define Roberts Cross kernels
kernel_x = np.array([[1, 0], [0, -1]], dtype=np.float64)
kernel_y = np.array([[0, 1], [-1, 0]], dtype=np.float64)

# 4. Apply kernels using convolution
roberts_x = cv2.filter2D(blurred, cv2.CV_64F, kernel_x)
roberts_y = cv2.filter2D(blurred, cv2.CV_64F, kernel_y)

# 5. Calculate Magnitude (Combine X and Y)
# Magnitude = sqrt(Gx^2 + Gy^2)
roberts_combined = cv2.magnitude(roberts_x, roberts_y)

# 6. Convert to uint8 for display
roberts_final = np.uint8(np.absolute(roberts_combined))

# Display
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1), plt.imshow(image, cmap='gray'), plt.title('Original Gray')
plt.subplot(1, 2, 2), plt.imshow(roberts_final, cmap='gray'), plt.title('Roberts Ed
plt.show()
```



Prewitt Edge Detector

In [4]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# 1. Load a real image in grayscale
img = cv2.imread(imgFile)

# 2. Grayscale & Blur (Essential for real photos to reduce grain)
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
blurred = cv2.GaussianBlur(gray, (5, 5), 0)

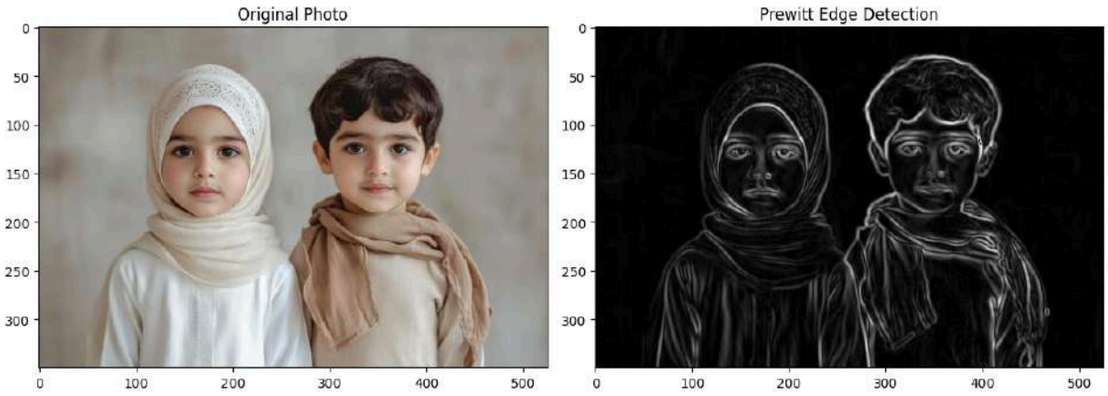
# 3. Define Prewitt Kernels manually
kernel_x = np.array([[-1, 0, 1], [-1, 0, 1], [-1, 0, 1]], dtype=np.float64)
kernel_y = np.array([[-1, -1, -1], [0, 0, 0], [1, 1, 1]], dtype=np.float64)

# 4. Apply filters
prewitt_x = cv2.filter2D(blurred, cv2.CV_64F, kernel_x)
prewitt_y = cv2.filter2D(blurred, cv2.CV_64F, kernel_y)

# 5. Calculate Magnitude (Combine X and Y)
prewitt_combined = cv2.magnitude(prewitt_x, prewitt_y)
```

```
# 6. Normalize to 8-bit for display
prewitt_final = np.uint8(np.absolute(prewitt_combined))

# Visualizing results
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1), plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB)), plt.title('
plt.subplot(1, 2, 2), plt.imshow(prewitt_final, cmap='gray'), plt.title('Prewitt Ed
plt.tight_layout()
plt.show()
```



Canny Edge Detector

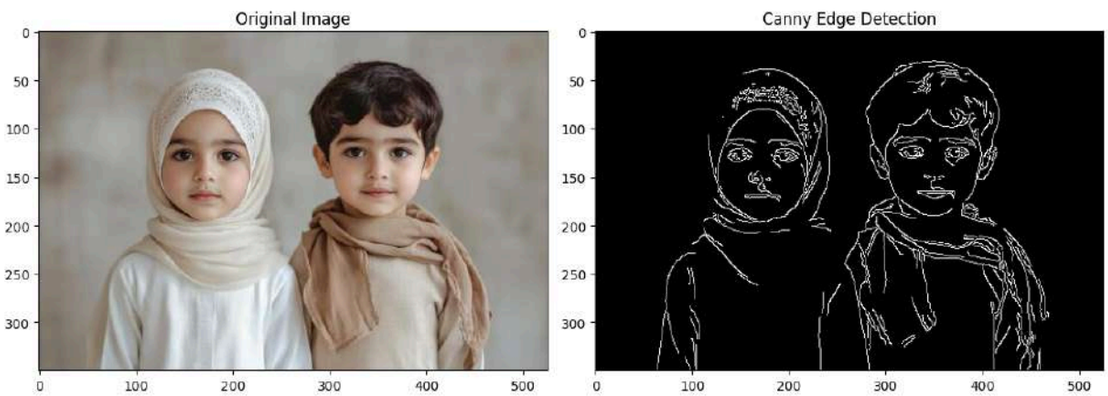
In [5]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# 1. Load the real image
# Replace 'building.jpg' with your file path
img = cv2.imread(imgFile)


# 2. Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# 3. Apply Canny Edge Detection
# Syntax: cv2.Canny(image, low_threshold, high_threshold)
# Tip: A 1:2 or 1:3 ratio for thresholds usually works best
canny_edges = cv2.Canny(gray, 100, 200)

# 4. Visualization
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1), plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB)), plt.title('
plt.subplot(1, 2, 2), plt.imshow(canny_edges, cmap='gray'), plt.title('Canny Edge D
plt.tight_layout()
plt.show()
```



Multi-Channel Canny Edge Detector

In [6]:
```
def best_color_edge_detector(image_path):
    # 1. Load the color image
    img = cv2.imread(image_path)
    if img is None:
        return "Image not found."
```

```
    # 2. Split into B, G, R channels
    channels = cv2.split(img)
    edge_maps = []

    for channel in channels:
        # 3. Pre-process each channel with Gaussian Blur to reduce noise
        blurred = cv2.GaussianBlur(channel, (5, 5), 0)

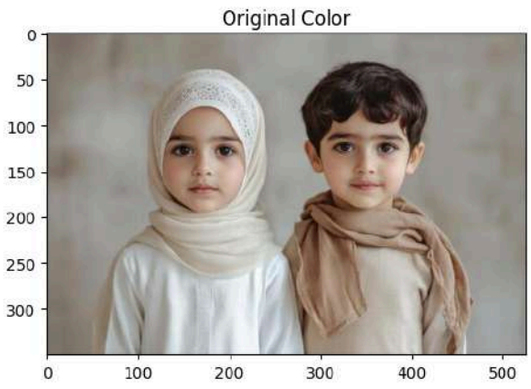
        # 4. Apply Canny to each channel
        # 50 and 150 are the low and high thresholds
        edges = cv2.Canny(blurred, 50, 150)
        edge_maps.append(edges)

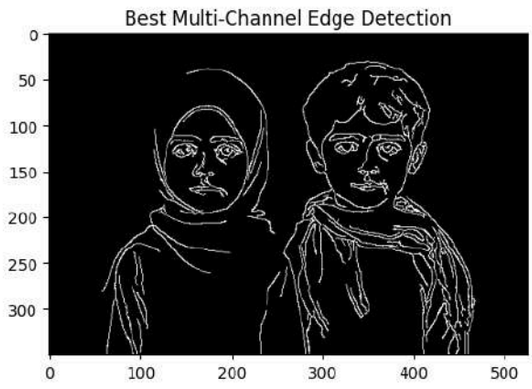
    # 5. Merge edges from all channels using bitwise OR
    # This keeps any edge found in ANY of the three color layers
    combined_edges = cv2.bitwise_or(edge_maps[0], edge_maps[1])
    combined_edges = cv2.bitwise_or(combined_edges, edge_maps[2])

    return img, combined_edges

# Run the detector
original, edges = best_color_edge_detector(imgFile)

# Display results
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1), plt.imshow(cv2.cvtColor(original, cv2.COLOR_BGR2RGB)), plt.ti
plt.subplot(1, 2, 2), plt.imshow(edges, cmap='gray'), plt.title('Best Multi-Channel
plt.show()
```





LoG Filter

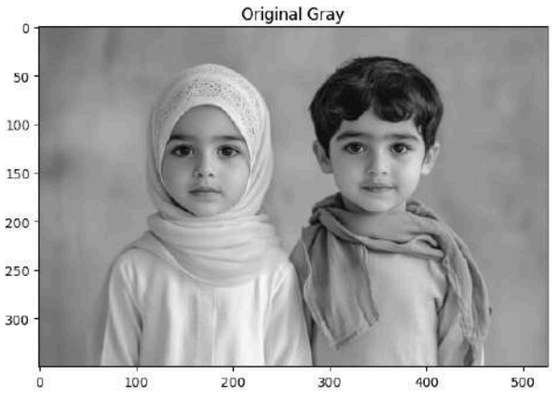
In [7]:
```
# 1. Load a real-world photo in grayscale
img = cv2.imread(imgFile, cv2.IMREAD_GRAYSCALE)

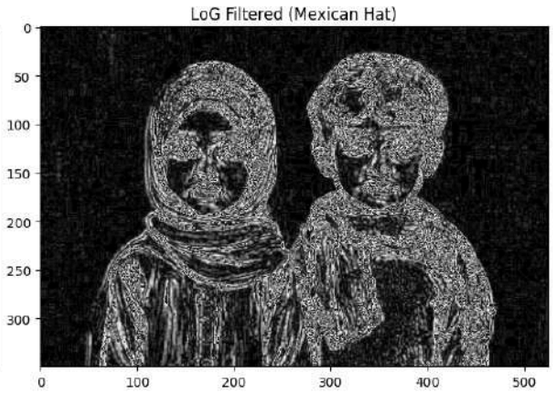
# 2. Step 1: Gaussian Blur (Smoothing)
# Larger kernel/sigma reduces more noise but may miss small edges
blurred = cv2.GaussianBlur(img, (5, 5), 0)

# 3. Step 2: Laplacian Operator (Second Derivative)
# cv2.CV_64F is used to store negative values (crucial for zero-crossings)
laplacian = cv2.Laplacian(blurred, cv2.CV_64F, ksize=5)

# 4. Post-processing: Normalize for display
# Since the Laplacian has negative and positive values, we take the absolute value
log_final = np.uint8(np.absolute(laplacian))

# Visualization
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1), plt.imshow(img, cmap='gray'), plt.title('Original Gray')
plt.subplot(1, 2, 2), plt.imshow(log_final, cmap='gray'), plt.title('LoG Filtered (
plt.tight_layout()
plt.show()
```





DoG Filter

In [8]:
```
# 1. Load the real-world photo
img = cv2.imread(imgFile, cv2.IMREAD_GRAYSCALE)

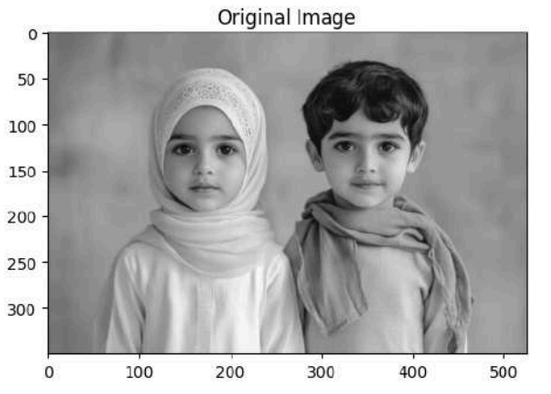
# 2. Apply two Gaussian blurs with different standard deviations (Sigmas)
# Low sigma preserves detail; High sigma captures broad structure
blur_narrow = cv2.GaussianBlur(img, (0, 0), sigmaX=1.0)
blur_wide = cv2.GaussianBlur(img, (0, 0), sigmaX=2.0)

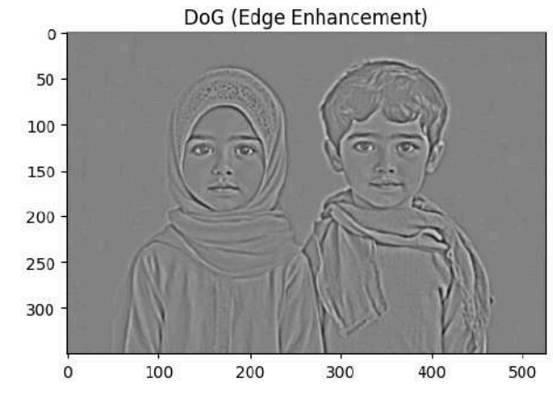
# 3. Calculate the Difference (DoG)
# Note: We use float32 to avoid overflow/underflow during subtraction
dog = blur_narrow.astype(np.float32) - blur_wide.astype(np.float32)

# 4. Normalize or Threshold for visibility
# Option A: Map to 0-255 range
dog_normalized = cv2.normalize(dog, None, 0, 255, cv2.NORM_MINMAX).astype(np.uint8)

# Option B: Simple threshold to create a binary edge map
_, dog_binary = cv2.threshold(dog_normalized, 127, 255, cv2.THRESH_BINARY)

# Visualization
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1), plt.imshow(img, cmap='gray'), plt.title('Original Image')
plt.subplot(1, 2, 2), plt.imshow(dog_normalized, cmap='gray'), plt.title('DoG (Edge
plt.show()
```





Gradient Edge Detector for Colored Images

In [9]:
```
# 1. Load the real-world color image
image = cv2.imread(imgFile)

# 2. Split the image into its R, G, and B components
# Note: OpenCV uses BGR order by default
b, g, r = cv2.split(image)

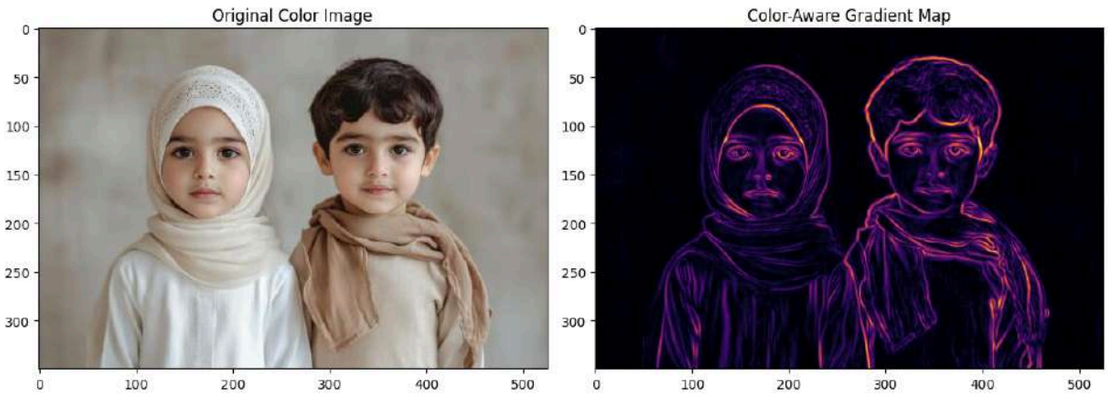
def get_channel_gradient(channel):
    # Use Sobel for gradient calculation on a single channel
    # Blur first to reduce noise in that specific color layer
    blurred = cv2.GaussianBlur(channel, (3, 3), 0)
    gx = cv2.Sobel(blurred, cv2.CV_64F, 1, 0, ksize=3)
    gy = cv2.Sobel(blurred, cv2.CV_64F, 0, 1, ksize=3)
    return cv2.magnitude(gx, gy)
```

```
# 3. Compute gradients for each color channel
grad_b = get_channel_gradient(b)
grad_g = get_channel_gradient(g)
grad_r = get_channel_gradient(r)

# 4. Combine gradients from all channels
# This ensures edges from color-only transitions are captured
combined_gradient = np.sqrt(grad_b**2 + grad_g**2 + grad_r**2)

# 5. Normalize to 0-255 for visualization
color_edges = cv2.normalize(combined_gradient, None, 0, 255, cv2.NORM_MINMAX).astyp

# Visualization
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1), plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB)), plt.title
plt.subplot(1, 2, 2), plt.imshow(color_edges, cmap='inferno'), plt.title('Color-Awa
plt.tight_layout()
plt.show()
```



The lab task

In [10]:
```
imgFile = "Butterfly.jpg"

img = cv2.imread(imgFile)
assert img is not None, "Image not found, check the file name and path"
print(img.shape)

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

```
(532, 800, 3)
```

In [11]:
```
# Blur first to reduce noise
blurred = cv2.GaussianBlur(gray, (5, 5), 0)

sobelx = cv2.Sobel(blurred, cv2.CV_64F, 1, 0, ksize=3)
sobely = cv2.Sobel(blurred, cv2.CV_64F, 0, 1, ksize=3)

sobel_combined = cv2.magnitude(sobelx, sobely)
sobel_final = np.uint8(np.absolute(sobel_combined))
```

In [12]:
```
kernel_x = np.array([[-1, 0, 1], [-1, 0, 1], [-1, 0, 1]], dtype=np.float64)
kernel_y = np.array([[-1, -1, -1], [0, 0, 0], [1, 1, 1]], dtype=np.float64)

prewitt_x = cv2.filter2D(blurred, cv2.CV_64F, kernel_x)
prewitt_y = cv2.filter2D(blurred, cv2.CV_64F, kernel_y)

prewitt_combined = cv2.magnitude(prewitt_x, prewitt_y)
prewitt_final = np.uint8(np.absolute(prewitt_combined))
```

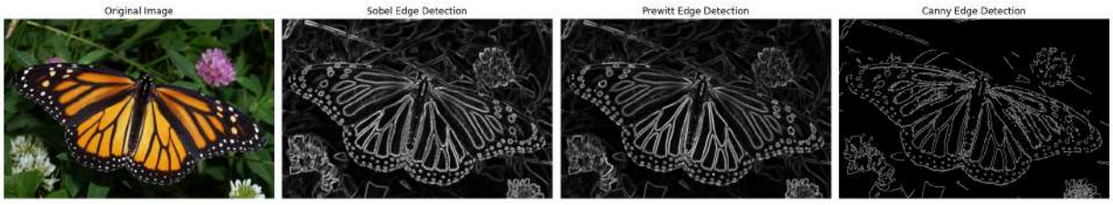
In [13]:
```
kernel_x = np.array([[-1, 0, 1], [-1, 0, 1], [-1, 0, 1]], dtype=np.float64)
kernel_y = np.array([[-1, -1, -1], [0, 0, 0], [1, 1, 1]], dtype=np.float64)

prewitt_x = cv2.filter2D(blurred, cv2.CV_64F, kernel_x)
prewitt_y = cv2.filter2D(blurred, cv2.CV_64F, kernel_y)
```

```
prewitt_combined = cv2.magnitude(prewitt_x, prewitt_y)
prewitt_final = np.uint8(np.absolute(prewitt_combined))
```

In [14]:
```
canny_edges = cv2.Canny(gray, 100, 200)
```

In [15]:
```
plt.figure(figsize=(20, 5))
plt.subplot(1, 4, 1), plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB)), plt.title('
plt.subplot(1, 4, 2), plt.imshow(sobel_final, cmap='gray'),   plt.title('Sobel Edge
plt.subplot(1, 4, 3), plt.imshow(prewitt_final, cmap='gray'), plt.title('Prewitt Ed
plt.subplot(1, 4, 4), plt.imshow(canny_edges, cmap='gray'),   plt.title('Canny Edge
plt.tight_layout()
plt.show()
```



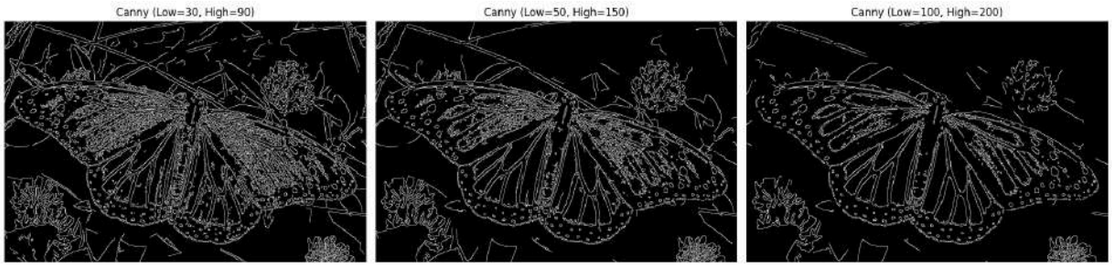
In [16]:
```
thresholds = [(30, 90), (50, 150), (100, 200)]

plt.figure(figsize=(18, 6))
for i, (low, high) in enumerate(thresholds):
    edges = cv2.Canny(gray, low, high)
    count = np.count_nonzero(edges)
    percent = 100 * count / edges.size
    print(f"Low={low}, High={high} -> edge pixels: {count} ({percent:.2f}%)")

    plt.subplot(1, 3, i + 1)
    plt.imshow(edges, cmap='gray')
    plt.title(f'Canny (Low={low}, High={high})')
    plt.axis('off')

plt.tight_layout()
plt.show()
```

```
Low=30, High=90 -> edge pixels: 53410 (12.55%)
Low=50, High=150 -> edge pixels: 38430 (9.03%)
Low=100, High=200 -> edge pixels: 26682 (6.27%)
```



Explanation

Raising the Canny thresholds reduces the number of detected edges and hides fine details, giving a cleaner but less detailed result. Lowering them detects more edges and shows more details, but also adds noise.